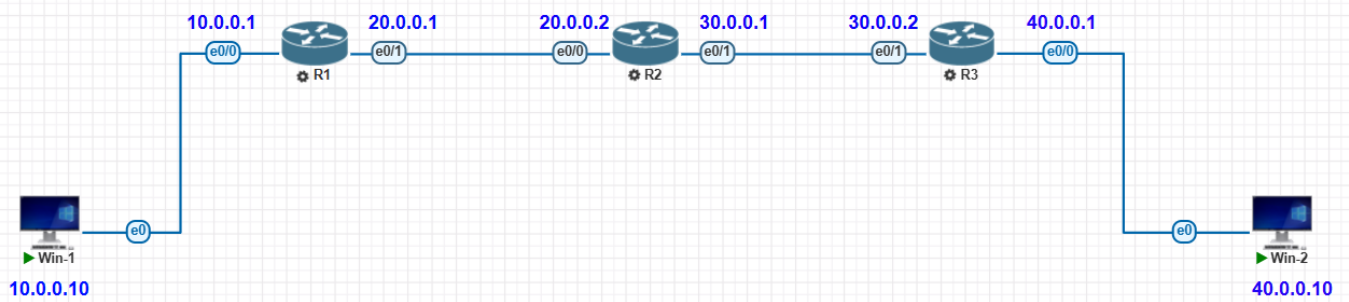
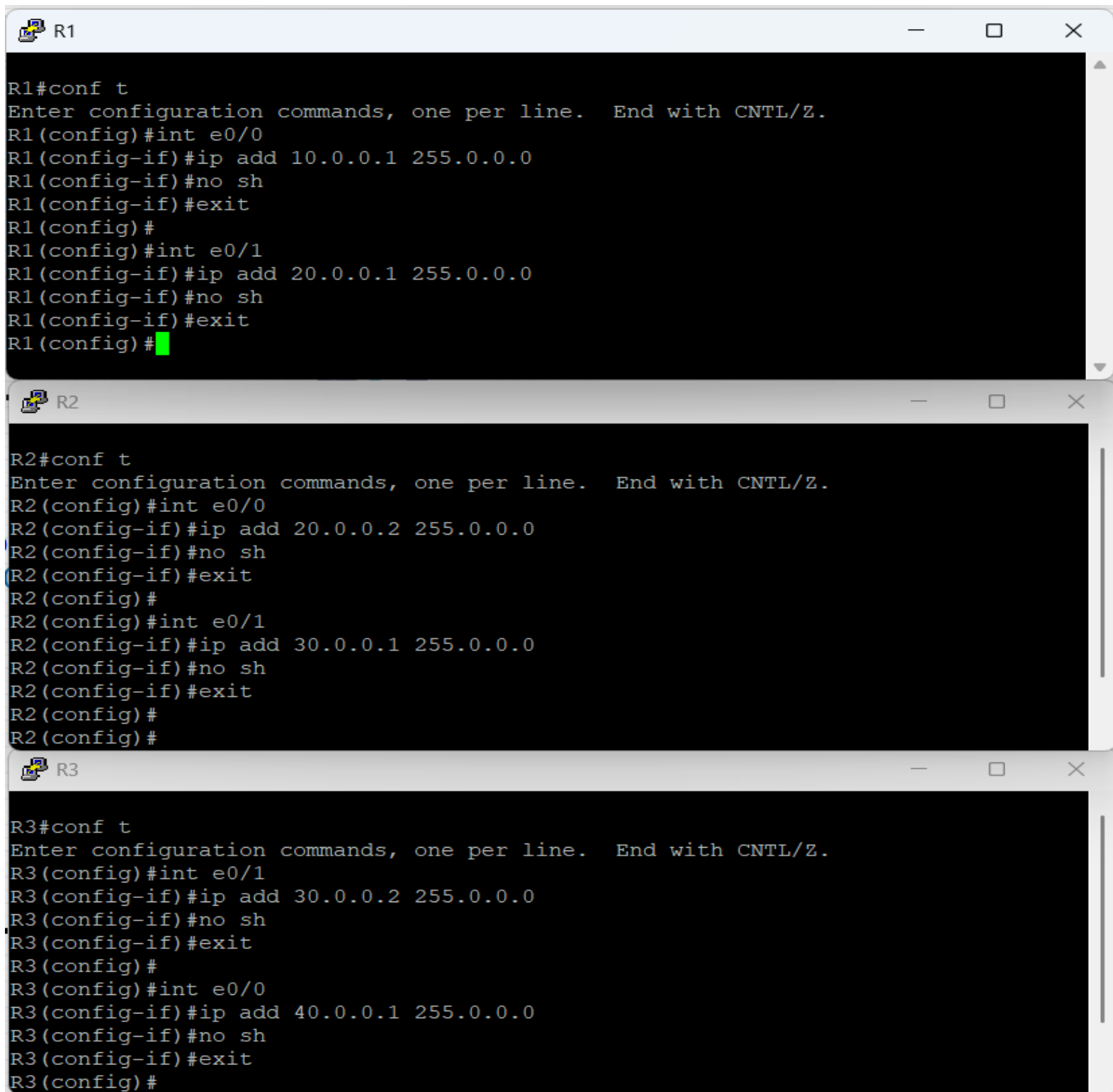


RIP CONFIGURATION



RIP

Step 1 : Assign ip address in all cisco routers



The image displays three terminal windows, each representing a Cisco router (R1, R2, and R3) in configuration mode. Each window shows the process of assigning IP addresses to interfaces e0/0 and e0/1. The R1 window shows configuration for 10.0.0.1 and 20.0.0.1. The R2 window shows configuration for 20.0.0.2 and 30.0.0.1. The R3 window shows configuration for 30.0.0.2 and 40.0.0.1. All configurations include the 'no sh' command to disable shutdown.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#

R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int e0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
R2(config)#int e0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
R2(config)#

R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int e0/1
R3(config-if)#ip add 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
R3(config)#int e0/0
R3(config-if)#ip add 40.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
```

Step 2 : Enable rip protocol in all cisco router

The image displays three overlapping terminal windows, each representing a Cisco router configuration session. The top window is for R1, the middle for R2, and the bottom for R3. Each window shows the command-line interface with the following commands entered:

```
R1(config)#  
R1(config)#router rip  
R1(config-router)#  
R1(config-router)#network 10.0.0.0  
R1(config-router)#network 20.0.0.0  
R1(config-router)#exit  
R1(config)#
```

```
R2(config)#  
R2(config)#router rip  
R2(config-router)#  
R2(config-router)#network 20.0.0.0  
R2(config-router)#network 30.0.0.0  
R2(config-router)#exit  
R2(config)#exit
```

```
R3(config)#  
R3(config)#router rip  
R3(config-router)#  
R3(config-router)#network 30.0.0.0  
R3(config-router)#network 40.0.0.0  
R3(config-router)#exit  
R3(config)#
```

Step 3 : Check routing table and calculate metric calculation

R1#sh ip route

```
*Sep 21 16:42:02.659: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
30.0.0.0/8 [120/1] via 20.0.0.2, 00:00:25, Ethernet0/1
40.0.0.0/8 [120/2] via 20.0.0.2, 00:00:25, Ethernet0/1
R1#
```

R2#sh ip route

```
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 20.0.0.1, 00:00:12, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
R    40.0.0.0/8 [120/1] via 30.0.0.2, 00:00:04, Ethernet0/1
R2#
```

R3#sh ip route

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

R    10.0.0.0/8 [120/2] via 30.0.0.1, 00:00:08, Ethernet0/1
R    20.0.0.0/8 [120/1] via 30.0.0.1, 00:00:08, Ethernet0/1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```

QEMU (Win-1)

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 0 . 0 . 10

Subnet mask: 255 . 0 . 0 . 0

Default gateway: 10 . 0 . 0 . 1

QEMU (Win-2)

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 40 . 0 . 0 . 10

Subnet mask: 255 . 0 . 0 . 0

Default gateway: 40 . 0 . 0 . 1

QEMU (Win-1)



C:\WINDOWS\system32\CMD.exe

Microsoft Windows XP [Version 5.1.2600]
<C> Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>PING 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=3ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125

Ping statistics for 40.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>_